


Started on Friday, 14 August 2026, 3:15 PM

State Finished

Completed on Friday, 14 August 2026, 3:44 PM

Time taken 28 mins 38 secs

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n;
6     int i = 1;
7     int s = 1;
8     int count = 0;
9     count++;
10
11
12     scanf("%d", &n);
13     count++;
14
15
```

```

        i++;
        s += i;
    }
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
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Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int i = 1;
7      int s = 1;
8      int count = 0;
9      count++;
10
11
12      scanf("%d", &n);
13      count++;
14
15
16      while (s <= n)
17      {
18          count++;
19          i++;
20          count++;
21          s += i;
22          count++;
23      }
24
25      count++;
26
27      printf("%d", count);
28
29      return 0;
30 }

```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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